What is Classification?

Classification is used when you want to predict a **category or class.**

The output is always **discrete** — it belongs to one of the defined classes.

Simple example:

- Email — spam or not spam
- Patient — disease or no disease
- Transaction — fraud or not fraud

Types of Classification

1️⃣ Binary Classification

Output has only **2 classes.**

Examples:

- Heart disease — yes or no
- Spam — spam or not spam
- Fraud — fraud or not fraud

2️⃣ Multiclass Classification

Output has **more than 2 classes.**

Examples:

- Classify handwritten digits — 0 to 9
- Classify news article — sports, politics, technology, entertainment
- Classify fruit — apple, banana, mango, orange

3️⃣ Multilabel Classification

Each input can belong to **multiple classes at same time.**

Examples:

- A movie can be action, comedy, and romance at same time
- A news article can be tagged with multiple topics
- A patient can have multiple diseases at same time

Classification Algorithms — What They Are

1️⃣ Logistic Regression

Despite the name it is a **classification** algorithm not regression.

It predicts the **probability** that an input belongs to a class. If probability is above 0.5 — class 1. Below 0.5 — class 0.

Uses a sigmoid function that converts any number into a value between 0 and 1.

Best for:

- Binary classification
- When you need probability scores
- Simple and fast baseline model
- When data is linearly separable

2️⃣ K-Nearest Neighbors — KNN

Classifies a new point based on the **majority class of its K nearest neighbors.**

Think of it like this — if you move to a new city and want to know what kind of neighborhood you are in, you look at your nearest neighbors. If 7 out of 10 nearest neighbors are doctors, you are probably in a doctors neighborhood.

How it works:

- Choose K — number of neighbors
- Calculate distance from new point to all training points
- Find K closest points
- Majority class among K neighbors becomes the prediction

Best for:

- Small datasets
- When decision boundary is complex
- Simple to understand and implement

Weakness:

- Slow on large datasets — calculates distance to every point
- Sensitive to irrelevant features
- K value selection is important

3️⃣ Decision Tree

Makes decisions by asking a **series of questions** about features — like a flowchart.

Think of it like playing 20 questions. You ask questions one by one and narrow down to the answer.

Example for heart disease:

Is age > 50?

Yes → Is cholesterol > 200?

Yes → High Risk

No → Medium Risk

No → Is blood pressure > 130?

Yes → Medium Risk

No → Low Risk

How it works:

- Starts at root node
- Splits data based on best feature at each step
- Continues until leaf node — final prediction
- Best split is chosen using Gini impurity or Information Gain

Best for:

- Easy to understand and visualize
- Works with both numerical and categorical data
- No need to scale features

Weakness:

- Overfits easily — grows too deep
- Small change in data can change entire tree

4️⃣ Random Forest

Collection of **many decision trees** working together.

Think of it like asking 100 doctors instead of 1. Each doctor gives their opinion and the majority vote wins.

How it works:

- Creates many decision trees — each trained on random subset of data
- Each tree gives a prediction
- Final prediction is majority vote of all trees
- This is called **ensemble learning**

Best for:

- Most reliable general purpose classifier
- Handles large datasets well
- Resistant to overfitting
- Works well without much tuning

Weakness:

- Slower than single decision tree

- Less interpretable — hard to explain why it made a decision

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5️⃣ Support Vector Machine — SVM

Finds the best **boundary line** that separates classes with maximum margin.

Think of it like drawing a line between two groups of points. SVM finds the line that has the maximum gap between both groups. Points closest to the line are called **support vectors.**

How it works:

- Finds optimal hyperplane that separates classes
- Maximizes margin between classes
- Uses kernel trick for non-linear data

Best for:

- High dimensional data
- When classes are clearly separable
- Text classification

Weakness:

- Slow on large datasets
- Hard to interpret
- Sensitive to feature scaling

6️⃣ Naive Bayes

Based on **probability and Bayes theorem.**

Assumes all features are independent of each other — which is the naive assumption.

Think of it like a doctor diagnosing a disease. They look at each symptom independently and calculate the probability of each disease based on those symptoms.

Best for:

- Text classification — spam detection, sentiment analysis
- Very fast and simple
- Works well with small data

Weakness:

- Assumption of independence is rarely true in real data
- Not great for complex relationships

7️⃣ XGBoost

One of the most powerful and popular algorithms in ML competitions.

Builds trees **sequentially** — each new tree corrects the errors of previous trees. This is called **boosting.**

Best for:

- Structured tabular data
- Kaggle competitions
- When you need highest accuracy
- Large datasets

Weakness:

- Many hyperparameters to tune
- Can overfit if not tuned properly
- Less interpretable

Key Evaluation Metrics for Classification

Metric	Meaning	When to use
Accuracy	Percentage of correct predictions	Balanced dataset
Precision	Of all predicted positive — how many were actually positive	When false positive is costly
Recall	Of all actual positive — how many did we catch	When false negative is costly
F1 Score	Balance between precision and recall	Imbalanced dataset
ROC AUC	How well model separates classes	Binary classification

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Precision vs Recall — Very Important

Precision — When model says positive, how often is it right?

Out of all patients predicted to have disease, how many actually have it?

Recall — Of all actual positives, how many did model catch?

Out of all patients who actually have disease, how many did model correctly identify?

Which matters more for heart disease? Recall is more important. Missing a sick patient (false negative) is more dangerous than incorrectly flagging a healthy patient (false positive).

Confusion Matrix — Know This

	Predicted No	Predicted Yes
Actual No	TN	FP
Actual Yes	FN	TP

- **TP** — True Positive — correctly predicted disease
- **TN** — True Negative — correctly predicted no disease
- **FP** — False Positive — predicted disease but patient is healthy
- **FN** — False Negative — predicted no disease but patient is sick

For medical problems **FN is most dangerous.**

Practice These

What is the difference between classification and regression?

"Classification predicts a discrete category like yes or no, while regression predicts a continuous numerical value like price or temperature. Classification uses metrics like accuracy, precision and recall. Regression uses metrics like R2 score and RMSE."

Why did you choose KNN for your heart disease project?

"I trained multiple models — Logistic Regression, Decision Tree, Random Forest and KNN. I compared them based on accuracy, precision and recall. Since this is a medical problem recall is more important than precision because missing a sick patient is dangerous. KNN gave the best recall score on my dataset so I chose it as the final model."

What is the difference between precision and recall?

"Precision measures how many of the predicted positives are actually positive. Recall measures how many of the actual positives the model correctly identified. In medical diagnosis recall is more critical because a false negative — missing a sick patient — is more dangerous than a false positive."

What is Random Forest?

"Random Forest is an ensemble of multiple decision trees. Each tree is trained on a random subset of data and features. The final prediction is the majority vote of all trees. It is more accurate and resistant to overfitting compared to a single decision tree."